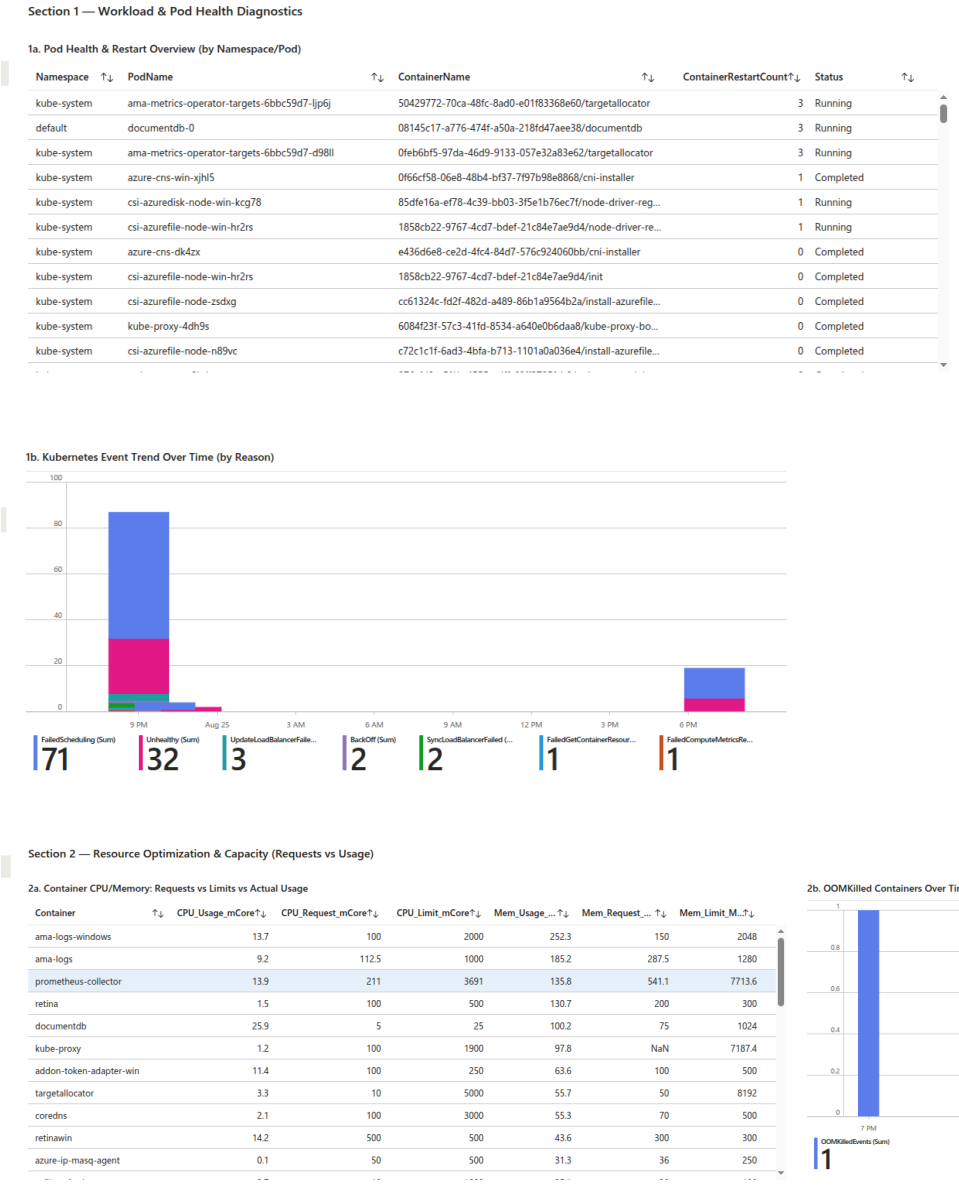
Azure AKS Observability & Monitoring

1. Azure Monitor Workbook

Here are the KQL queries for cluster events and metrics:



Section 1 — Workload & Pod Health Diagnostics

1a. Pod Health & Restart Overview (by Namespace/Pod)

Namespace	PodName	ContainerName	ContainerRestartCount	Status
kube-system	ama-metrics-operator-targets-6bbc59d7-ljp6j	50429772-70ca-48fc-8ad0-e01f83368e60/targetallocator	3	Running
default	documentdb-0	08145c17-a776-474f-a50a-218fd47aee38/documentdb	3	Running
kube-system	ama-metrics-operator-targets-6bbc59d7-d98ll	0feb6bf5-97da-46d9-9133-057e32a83e62/targetallocator	3	Running
kube-system	azure-cns-win-xjhl5	0f66cf58-06e8-48b4-bf37-7f97b98e8868/cni-installer	1	Completed
kube-system	csi-azuredisk-node-win-kcg78	85dfe16a-ef78-4c39-bb03-3f5e1b76ec7f/node-driver-reg...	1	Running
kube-system	csi-azurefile-node-win-hr2rs	1858cb22-9767-4cd7-bdef-21c84e7ae9d4/node-driver-re...	1	Running
kube-system	azure-cns-dk4zx	e436d6e8-ce2d-4fc4-84d7-576c924060bb/cni-installer	0	Completed
kube-system	csi-azurefile-node-win-hr2rs	1858cb22-9767-4cd7-bdef-21c84e7ae9d4/init	0	Completed
kube-system	csi-azurefile-node-zsdxg	cc61324c-fd2f-482d-a489-86b1a9564b2a/install-azurefile...	0	Completed
kube-system	kube-proxy-4dh9s	6084f23f-57c3-41fd-8534-a640e0b6daa8/kube-proxy-bo...	0	Completed
kube-system	csi-azurefile-node-n89vc	c72c1c1f-6ad3-4bfa-b713-1101a0a036e4/install-azurefile...	0	Completed

1b. Kubernetes Event Trend Over Time (by Reason)

Section 2 — Resource Optimization & Capacity (Requests vs Usage)

2a. Container CPU/Memory: Requests vs Limits vs Actual Usage

Container	CPU_Usage_mCore	CPU_Request_mCore	CPU_Limit_mCore	Mem_Usage_...	Mem_Request_...	Mem_Limit_M...
ama-logs-windows	13.7	100	2000	252.3	150	2048
ama-logs	9.2	112.5	1000	185.2	287.5	1280
prometheus-collector	13.9	211	3691	135.8	541.1	7713.6
retina	1.5	100	500	130.7	200	300
documentdb	25.9	5	25	100.2	75	1024
kube-proxy	1.2	100	1900	97.8	NaN	7187.4
addon-token-adapter-win	11.4	100	250	63.6	100	500
targetallocator	3.3	10	5000	55.7	50	8192
coredns	2.1	100	3000	55.3	70	500
retinawin	14.2	500	500	43.6	300	300
azure-ip-masq-agent	0.1	50	500	31.3	36	250

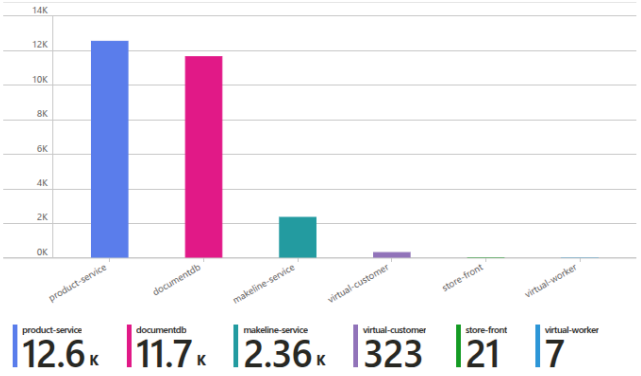
2b. OOMKilled Containers Over Time

Section 3 — Container & Application Logs Analysis

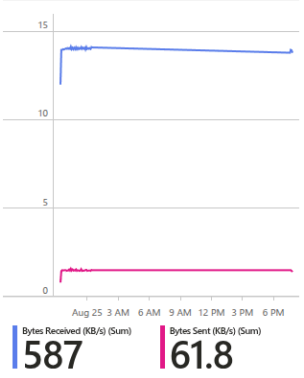
3a. Container Error & Warning Logs (stderr / Error / Exception / Pino level >=50)

TimeGenerated	Computer	ContainerID	CleanLog
8/25/2026, 7:51:37.315 PM	aks-lxpool-66708481-vmss000000	fd83cfc998ddb9cc9b34f41ed3952c11d9111775d711ba0d...	2026/08/25 15:51:37 AMQP consumer error: failed to con...
8/25/2026, 7:51:32.314 PM	aks-lxpool-66708481-vmss000000	fd83cfc998ddb9cc9b34f41ed3952c11d9111775d711ba0d...	2026/08/25 15:51:32 AMQP consumer error: failed to con...
8/25/2026, 7:51:27.312 PM	aks-lxpool-66708481-vmss000000	fd83cfc998ddb9cc9b34f41ed3952c11d9111775d711ba0d...	2026/08/25 15:51:27 AMQP consumer error: failed to con...
8/25/2026, 7:51:22.310 PM	aks-lxpool-66708481-vmss000000	fd83cfc998ddb9cc9b34f41ed3952c11d9111775d711ba0d...	2026/08/25 15:51:22 AMQP consumer error: failed to con...
8/25/2026, 7:51:17.309 PM	aks-lxpool-66708481-vmss000000	fd83cfc998ddb9cc9b34f41ed3952c11d9111775d711ba0d...	2026/08/25 15:51:17 AMQP consumer error: failed to con...
8/25/2026, 7:51:12.306 PM	aks-lxpool-66708481-vmss000000	fd83cfc998ddb9cc9b34f41ed3952c11d9111775d711ba0d...	2026/08/25 15:51:12 AMQP consumer error: failed to con...
8/25/2026, 7:51:07.945 PM	aks-lxpool-66708481-vmss000000	fd83cfc998ddb9cc9b34f41ed3952c11d9111775d711ba0d...	2026/08/25 15:51:07 Failed to get pending orders from d...
8/25/2026, 7:51:07.945 PM	aks-lxpool-66708481-vmss000000	fd83cfc998ddb9cc9b34f41ed3952c11d9111775d711ba0d...	2026/08/25 15:51:07 Failed to find records: server selectio...
8/25/2026, 7:51:07.944 PM	aks-lxpool-66708481-vmss000000	7322ac0ee5d2a95ff04df23af58d7c566a7b45a9f0a0b1b6e...	Failed to fetch orders: error sending request for url (http:/...
8/25/2026, 7:51:07.205 PM	aks-lxpool-66708481-vmss000000	fd83cfc998ddb9cc9b34f41ed3952c11d9111775d711ba0d...	2026/08/25 15:51:07 AMQP consumer error: failed to con...
8/25/2026, 7:51:06.928 PM	aks-lxpool-66708481-vmss000000	a07e184b9b5e841390d5779af7514943934 5f571d063efb0...	Failed to submit order: error sending request for url (http:...

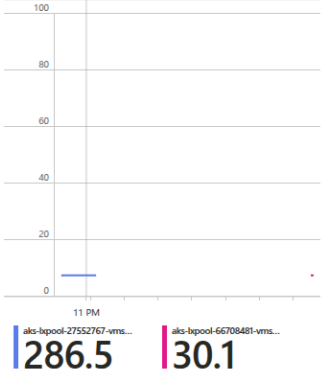
3b. Top 10 Error-Generating Containers



4a. Network I/O by Node (Bytes Received / Sent)

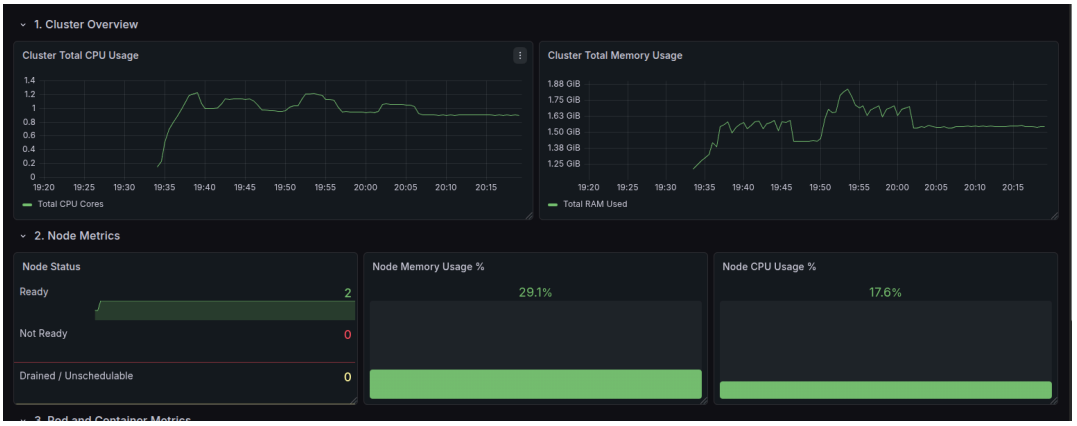


4b. Node Disk Usage (%)

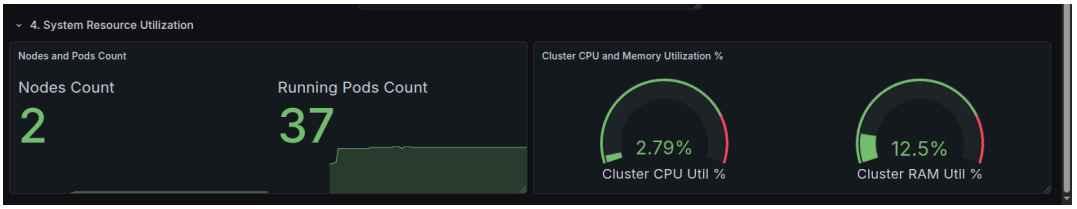


2. Grafana Monitoring Dashboard

Custom Grafana dashboard for AKS metrics:

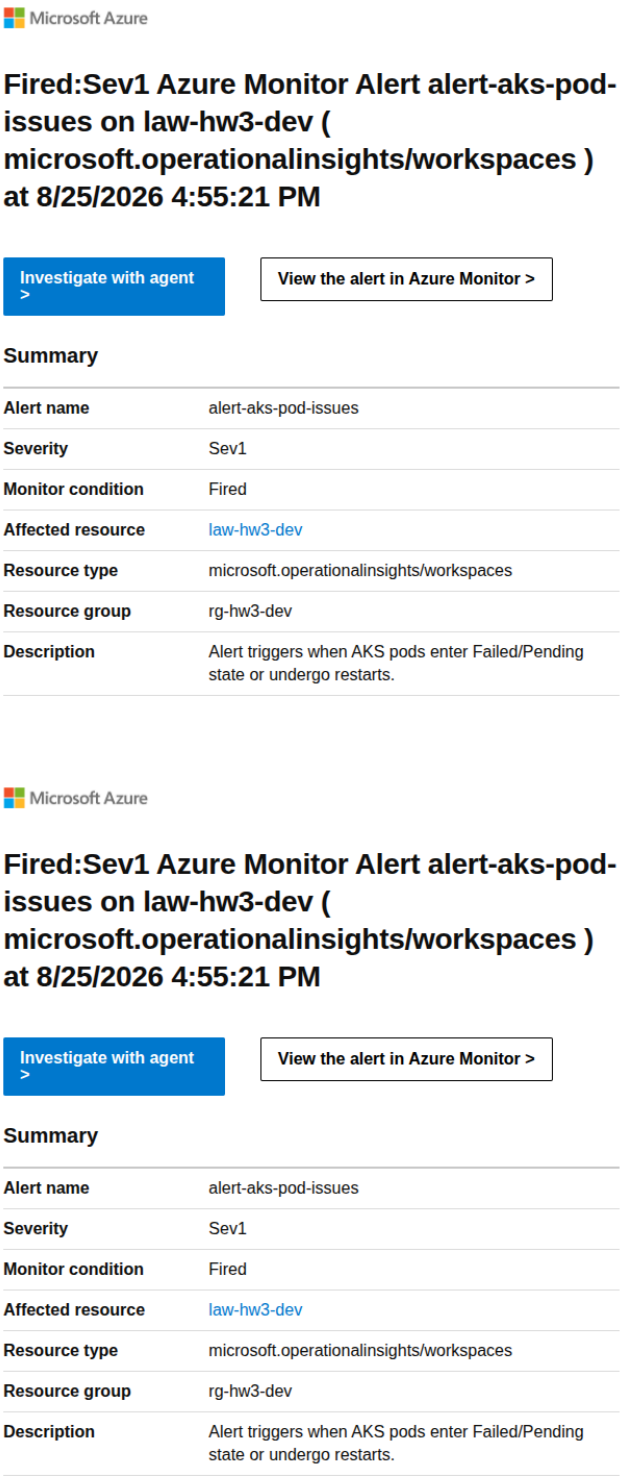






2. Triggered Email Alerts

Email alert notification triggered by Azure Monitor:



Microsoft Azure
Fired:Sev1 Azure Monitor Alert alert-aks-pod-issues on law-hw3-dev (microsoft.operationalinsights/workspaces) at 8/25/2026 4:55:21 PM
Investigate with agent >
View the alert in Azure Monitor >
Summary
Alert name alert-aks-pod-issues
Severity Sev1
Monitor condition Fired
Affected resource law-hw3-dev
Resource type microsoft.operationalinsights/workspaces
Resource group rg-hw3-dev
Description Alert triggers when AKS pods enter Failed/Pending state or undergo restarts.

---

4. Azure Native Dashboard

Azure Native Dashboard:


Auto refresh : Off
UTC Time : Past 24 hours
Add filter
Cluster Node Readiness (Ready vs NotReady)
Ready (100%)
Ready
Node CPU Utilization (%)
CPU Usage (%)
75
50
25
0
6:00 PM
Aug 25
6:00 AM
12:00 PM
TimeGenerated
aks-lxpool-27552767-vmss000000
aks-lxpool-66708481-vmss000000
akswin000000
Node Memory Working Set (%)
Memory Usage (%)
50
40
30
20
10
6:00 PM
Aug 25
6:00 AM
12:00 PM
TimeGenerated
aks-lxpool-27552767-vmss000000
aks-lxpool-66708481-vmss000000
akswin000000
Pod Count by Status (Running / Pending / Failed)
Pod Count
40
30
20
10
0
Running
Pending
Phase
PodCount
Active Node Count by Node Pool
Active Nodes
1.25
1
0.75
0.5
0.25
0
lxpool
win
Node Pool
ActiveNodes